

Roll No.....

ODD END SEMESTER EXAMINATION DECEMBER - 2025

Name of the Program: **B. Tech. Civil Engineering**

Semester: **V**

Name of the Course: **General Geology**

Course Code: **TPE 302**

Time: **3 Hours**

Maximum Marks: **100**

Note:

- (i) All questions are compulsory.
- (ii) Answer any **two** sub questions among **a, b & c** in each main question.
- (iii) Total marks for each main question are **twenty**.
- (iv) Each question carries **10 Marks**

Q1	(20 marks)	
(a)	Describe the types of plate boundaries and the engineering hazards associated with each.	CO1
(b)	Describe the internal structure of the Earth with neat sketches and explain its relevance to civil engineering.	
(c)	Explain how earthquakes are measured and how seismic data are interpreted for design.	
Q2	(20 marks)	
(a)	Define mass wasting and discuss the factors controlling slope stability.	CO1 CO2
(b)	Differentiate between i) Streak and Colour ii) Cleavage and Fracture.	
(c)	Discuss the effect of weathering on building stones and construction materials.	
Q3	(20 marks)	
(a)	Define a mineral and explain the physical properties used for mineral identification.	CO3
(b)	Explain the classification system and the significance of sedimentary rocks.	
(c)	Discuss the common rock types encountered in India and their engineering uses.	
Q4	(20 marks)	
(a)	Define stratigraphy and explain its importance in civil engineering.	CO4 CO5
(b)	Describe the Laws of Stratigraphy and their significance.	
(c)	Define fossils and describe the different types of fossils.	
Q5	(20 marks)	
(a)	Explain the formation of folds and outline the different types, using clear sketches.	CO5
(b)	What are faults in geology? Explain their various types with the help of neat diagrams.	
(c)	Discuss the applications of structural geology in civil engineering, particularly in site selection and hazard mitigation.	

